

Ablanvi Songo

Research Interests

My research is in the field of nonlinear analysis. I am particularly interested in linking theorems (for single-valued and multivalued functionals) in critical point theory, as well as their applications to the study of solutions to PDEs and partial differential inclusions using variational and topological methods.

Education

- Winter 2021 - **Ph.D. in Mathematics**, *Université de Sherbrooke, Canada*,
Summer 2026 Supervisors: Fabrice Colin, Vasilisa Shramchenko
- 2019-2020 **M.Sc. (1st year) in Actuarial and Financial Mathematics**, *Université de Rouen Normandie, France*
- 2018-2019 **M.Sc. (2nd year) in Fundamental and Applied Mathematics**, *Université de Rouen Normandie, France*
- 2017-2018 **M.Sc. (1st year) in Fundamental Mathematics**, *Aix Marseille Université, France*
- 2011-2015 **B.Sc. in Mathematics**, *Université de Lomé, Togo*

Publications and Preprints

Publications

- 2026 **Existence result for a system of two semilinear coupled Poisson equations with asymptotically linear nonlinearities**, *Differential Equations & Applications*, in Press
- 2026 **An infinite dimensional Saddle Point Theorem and application**, with F. Colin, *Boundary Value Problems*, (2026), <https://doi.org/10.1186/s13661-026-02234-8>

Preprints

- 2026+ **Mountain Pass Theorem for multivalued functionals and application to subcritical elliptic differential inclusion systems in R^2** , in preparation
- 2026+ **Linking theorems for multivalued functionals and application to partial differential inclusions**, in preparation, with F. Colin
- 2025 **An infinite-dimensional mountain pass theorem with applications to nonlinear elliptic systems**, with F. Colin, <https://doi.org/10.48550/arXiv.2512.02229>
- 2025 **On the Fountain Theorem for Continuous Functionals and Its Application to a Semilinear Elliptic Problem in R^2** , with F. Colin, <https://doi.org/10.48550/arXiv.2509.16059>
- 2025 **A new infinite-dimensional Linking theorem with application to a system of coupled Poisson equations**, with F. Colin, <https://doi.org/10.48550/arXiv.2506.17563>

Talks

- May 2026 **XXVIIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, UTQR, Trois-Rivières
- April 2026 **Présentations "180 secondes"**, Caféfak, Faculté des sciences, Université de Sherbrooke
- November 2025 **Séminaire Junior en Géométrie et Physique Mathématique**, Département de mathématiques, Université de Sherbrooke
- May 2025 **XXVIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, McGill University
- February 2025 **Club Math**, Department of Mathematics, Université de Sherbrooke
- Nov 2023 **Séminaire "Séminaire" d'étudiants gradués**, Department of Mathematics, Université de Sherbrooke

Awards and Grants

- 2021-2024 **Doctoral scholarship of the Faculty of Sciences**, Faculté des sciences, Université de Sherbrooke

- 2021-2026 **ISM Travel Grant**, *Institut des Sciences Mathématiques (ISM), Montréal*
- Hiver 2026 **Award for Outstanding First Publication**, *Faculté des sciences, Université de Sherbrooke*
- 2023-2024, 2022-2023 **ISM Graduate Scholarship**, *Institut des Sciences Mathématiques (ISM), Montréal*
- 2022-2023 **Bourse In memorian Ernest-Monga for perseverance and commitment**, *Département of Mathematics, Université de Sherbrooke*
- 2018-2019, 2017-2018 **Scholarship of Government of Togo**, *Ministère de l'Enseignement Supérieur et de la Recherche*

Conferences and Schools Attended

- May 2026 **XXVIIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *UTQR, Trois-Rivières*
- November 2025 **Séminaire Junior en Géométrie et Physique Mathématique**, *Département de mathématiques, Université de Sherbrooke*
- October 2025 **3rd University of North Carolina at Greensboro Virtual PDE Conference : dedication to Professor Alfonso Castro in celebration of his 75th birthday and his outstanding contributions to Differential Equations**, *Département of Mathematics and Statistics (online)*
- May 2025 **XXVIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *McGill University*
- June 2024 **XXVIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université de Québec à Montréal*
- October 2023 **Black Heroes of Mathematics Conference 2023**, *ICMS, the University of Edinburgh (online)*
- June 2023 **XXVe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université de Sherbrooke*
- July 2022 **Topological Methods in Nonlinear Analysis: Recent Advances - Conference in memory of Professor Andrzej Granas**, *Centre de recherches mathématiques (CRM), Université de Montréal*
- May 2022 **XXIVe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université Laval*

Teaching Experience

Université de Sherbrooke

- Winter 2026 **Teaching Assistant**, *MAT 117 (Differential and Integral Calculus of Several Variables and Vector Analysis)*
- Fall 2025 **Teaching Assistant**, *MAT 098 (Supplementary Material on Integral Calculus)*
- Summer 2025 **Teaching Assistant**, *MAT 455 (Analysis III - Differential and Integral Calculus in $\mathbb{R}^n$)*
- Winter 2025 **Teaching Assistant**, *MAT 129 (Analysis I - Elements of Analysis)*
- Fall 2024 **Teaching Assistant**, *MAT 199 (Linear Algebra Applied to Computer Science), MAT 221, CQP 208*
- Winter 2024 **Teaching Assistant**, *MAT 129, MAT 221 (Differential and Integral Calculus), CQP 208*
- Fall 2023 **Teaching Assistant**, *MAT 221 (Differential and Integral Calculus)*
- Winter 2023 **Teaching Assistant**, *MAT 129 (Elements of Analysis)*
- Fall 2022 **Teaching Assistant**, *MAT 221 (Differential and Integral Calculus), MAT 324*
- Summer 2022 **Teaching Assistant**, *MAT 455 (Analysis III)*
- Winter 2022 **Teaching Assistant**, *MAT 193 (Linear Algebra)*
- Fall 2021 **Teaching Assistant**, *MAT 324 (Mathematical Models)*
- Winter 2021 **Teaching Assistant**, *MAT 193 (Linear Algebra)*

Lycée de Davié (Togo)

- 2014-2017 **Nationally certified teacher**, *Mathematics, Physics-Chemistry*

Academic and Community Involvement

- March 2026 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Department**, *Sherbrooke*
- May 2025 **Volunteer during the final round of the International Mathematical and Logical Games Championship (province of Quebec)**, *Cegep de Sherbrooke*
- Mars 2025 **Volunteer during the semi-final round of the International Mathematical and Logical Games Championship (Sherbrooke Region)**, *Department of Mathematics, Université de Sherbrooke*

- March 2025 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Department, Sherbrooke**
- March 2024 **Volunteer during the semi-final round of the International Mathematical and Logical Games Championship (Sherbrooke Region), Department of Mathematics, Université de Sherbrooke**
- March 2024 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Département, Sherbrooke**
- May 2023 **Volunteer during the Quebec final of the International Mathematical and Logical Games Championship, Department of Mathematics, Université de Sherbrooke**
- Winter 2023 **Volunteer during the Faculty of Science Open House, Université de Sherbrooke**
- Fall 2022 **Volunteer during the Faculty of Science Open House, Université de Sherbrooke**
- 2021–2023 **Volunteer mathematics tutor for Le Diplôme Avant la Médaille (DAM) academic support program for student-athletes**

Technical Strengths and Skills

Programming

Python, $\text{\LaTeX}$

Languages

French, English, Ewè

Academic References

Fabrice Colin, *Associate Professor*, Laurentian University

Email: fcolin@laurentian.ca

Vasilisa Shramchenko, *Full Professor*, Université de Sherbrooke

Email: Vasilisa.Shramchenko@USherbrooke.ca